BEST2: validation remains open

27 October 2026 | PR #2689 | Accounting and correctness audit

BEST2 stays disabled. The historical 0.947 estimate is derived from a standalone iteration model and GPU throughput. It is not a measured GPU solve-cost ratio or an end-to-end 95% confidence interval.

Source findings and corrections

Timing repetitions reused identical starts and tables. The revised driver exposes reproducible seed namespaces and a fixed-table / varying-start control.

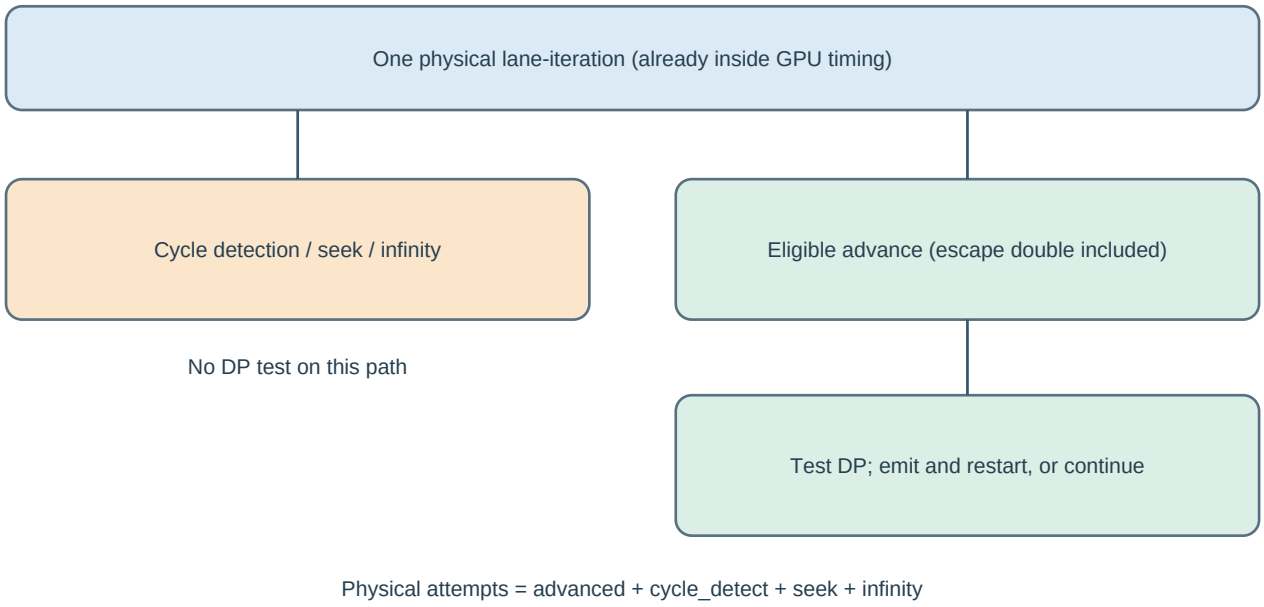
The DP predicate and cycle hash both use limb 0. Score limb 1 being different does not establish independence. Cycle-detection and seek iterations skip DP testing.

Warmup DPs were counted against a measured-only denominator. The driver now clears the counter after warmup; four warmup trajectory steps remain.

Host seed replay could lose the restart number after an infinity start. The caller now receives the actual restart; a constructed cancellation-start regression passes.

The standalone model uses hashed predicates, different escape mechanics and no restart after each DP. Its iteration ratio does not directly price the CUDA walk.

Count physical work and DP opportunities separately



Executed checks and remaining work

Small development tests establish correctness within their stated scope

Checks actually executed

Both CPU field / curve / rho suites pass for BEST2=0 and BEST2=1. Runtime-W and compile-time-W equivalence now includes W=64.

OpenSSL independently computes P+M and P-M on the named P-256 and P-384 curves: 2048 physical steps per arm, two namespaces, W=16, 32 iterations. Batch states and emitted DP identities match the per-phase trajectory.

Toy scalar replay checks 4000 phases per arm, including subtraction, detection/seek and escape doubling. An infinity-start regression verifies matching host/device restart identities.

The real Rust model source passes a dependency-free bookkeeping test: 12 seeds on both arms, correct synthetic group recoveries and identical replay counters. This is not a root integration build.

The summary tool reproduces a synthetic eight-seed fixture ratio of 1.1 and rejects duplicated seeds. These fixtures are not measured DP distributions.

What has not been established

The DP bias-versus-deterministic-sampling diagnosis remains unresolved. No statistically powered exact-kernel seeded panel ran. Unit-test DP totals are too small for that inference.

No authenticated GPU/EKS execution path, nvcc or device was available. No cloud job was submitted. Cargo 1.75 cannot parse the repository lockfile v4, so root build/session/replay evidence is missing.

The new ecbench model uses hashed group keys and charges two full additions. It is explicitly separate from the coordinate-limb CUDA walk and cannot validate the NIST GPU cost estimate.

A bounded known-answer GPU collision workload, GPU-to-host DP trace, independent scalar recovery and all-phase successful-collision accounting are still missing. The current RhoHost supports orders below 2^63 only.

Next discriminating experiments

The frozen protocol retains W=16/32/64 and both curves, fixed-table and varied-table panels, paired seed-cluster bootstrap uncertainty, failures and raw hashes. Timing repetitions are controls, not new samples. Require independent replay and number recomputation before a benchmark row or default enablement.

Primary-paper retrieval for Bernstein, Lange and Schwabe, eprint 2011/003, was blocked. This audit makes no paper-derived quantitative or novelty claim. See README.md for exact source paths, requirement status and blockers.